

What is diabetes

Diabetes mellitus, or diabetes, is a condition where there is too much glucose in the blood. The body can't make insulin, enough insulin or is not effectively using the insulin it does make. Over time high glucose levels can damage blood vessels and nerves, resulting in long term health complications including heart, kidney, eye and foot damage.

Diabetes is a serious and chronic condition which can affect the entire body. While there is currently no cure, you can live an enjoyable life by learning about the condition and effectively managing it.

There are three main types of diabetes; all types are complex and serious:

- [Type 1](#)
- [Type 2](#)
- [Gestational diabetes](#)

Pre-diabetes may be diagnosed before it progresses to type 2 diabetes.

How does diabetes affect the body?

When someone has diabetes, their [blood glucose levels](#) are too high. Glucose is a form of sugar which is the main source of energy for our bodies. High levels of glucose in the blood can lead to long term and short term health [complications](#).

Everything our bodies do requires energy. For our bodies to work properly we need to convert carbohydrates in food from starch and sugars into glucose which enters the bloodstream and is transported to the body's cells. If we don't have enough carbohydrates our body will make glucose from fat and protein.

A hormone called insulin is produced by the pancreas. It acts like a 'key' in the 'lock' of our cell's 'doors' to let the glucose in where it is used to provide energy.

In people with diabetes, depending on the type of diabetes, insulin is either no longer produced or not enough is produced or the body doesn't use it as well as it used to. When people with diabetes eat foods that contain carbohydrates, the glucose cannot enter the body's cells as the cell 'doors' don't open. The glucose stays in the blood resulting in high blood glucose levels.

In the short term, high levels of glucose in your blood make you feel tired, the kidneys work hard to pass some of the excess glucose out through the urine so you may go to the toilet to pass urine more often, and this can lead to dehydration and feeling thirsty.

In the long term, high blood glucose levels can damage small and large blood vessels and nerves which can cause damage your heart, brain, kidneys, eyes and feet.

Glucose levels can be monitored using a blood glucose monitor or continuous glucose monitor. Blood glucose levels are managed through self-care and treatment.

Three things you need to know about diabetes:

- It is not one condition – there are three main types of diabetes: [type 1](#), [type 2](#) and [gestational diabetes](#)
- All types of diabetes are complex and require daily care and [management](#)
- Diabetes does not discriminate, anyone can develop diabetes

Symptoms

In type 1 diabetes, symptoms are often sudden and can be life-threatening so it is usually diagnosed within hours or days.

In type 2 diabetes, many people have no symptoms at all, while other signs can go unnoticed being seen as part of 'getting older'. By the time symptoms are noticed, complications of diabetes may already be present.

Common symptoms include:

- Being more thirsty than usual

- Passing more urine
- Feeling tired and lethargic
- Always feeling hungry
- Having cuts that heal slowly
- Itching, skin infections
- Urinary tract or fungal infections
- Blurred vision
- Unexplained weight loss (type 1 or type 2 diabetes)
- Gradually putting on weight (type 2 diabetes)
- Mood swings
- Headaches
- Feeling dizzy
- Leg cramps

Note: This information is of a general nature only and should not be substituted for medical advice or used to alter medical therapy. It does not replace consultations with qualified healthcare professionals to meet your individual medical needs.

Diagnosis

Diabetes can only be [diagnosed](#) with a laboratory blood test ordered by your general practitioner or nurse practitioner. A finger prick test using a hand-held glucose meter cannot be used to diagnose diabetes.

There are different blood tests to diagnose diabetes¹:

1. Glycated haemoglobin (HbA1c): this non-fasting test is a measure of your **average** blood glucose over the last two to three months. It is measured in percent (%) or mmol/mol (millimoles per mole). An HbA1c of 6.5% (48 mmol/mol) or more means diabetes is likely. Between 6.0% to 6.4% would indicate pre-diabetes.
2. Fasting blood glucose: this test measures how much glucose is in your bloodstream at a certain time of day *when you haven't eaten for eight hours such as overnight*. It is measured in mmol/L (millimoles per litre). A fasting blood glucose of 7.0 mmol/L or more would indicate diabetes is likely. Between 6.1 mmol/L and 6.9 mmol/L is likely pre-diabetes (also called *impaired fasting glucose*).

3. Non-fasting (or random) blood glucose: this test measures how much glucose is in your bloodstream at a certain type of day *if you haven't been fasting* or have eaten something recently. It is measured in mmol/L (millimoles per litre). A random (non-fasting) blood glucose of 11.1 mmol/L or more would indicate diabetes. Between 7.8 mmol/L and 11.0 mmol/L is likely pre-diabetes (also called *impaired glucose tolerance*).
4. Oral glucose tolerance test (OGTT): this test combines a fasting and a non-fasting blood glucose test. After the fasting blood test, you will take a very sweet drink and your blood glucose will be tested at 1 hour and 2 hours. The results are in mmol/L (millimoles per litre). Diabetes is likely if your fasting level is 7.0mmol/L or above, or your 2-hour level is 11.1 mmol/L or above.

Your general practitioner may order a second blood test to confirm diabetes if you do not have symptoms.

Type 1 diabetes is usually confirmed with an extra blood test to check *autoantibodies*. These are the 'defence' proteins the immune system produces, which can stop the pancreas from making insulin. There are five known *autoantibodies* related to type 1 diabetes. If any of these are high, then that would confirm you have type 1 diabetes.

Diabetes is serious

Diabetes can be managed well but the potential complications are the same for type 1 and type 2 diabetes.

Early diagnosis and optimal self-management and treatment have been shown to reduce the risk of diabetes-related complications.

We know diabetes:

- Is the leading cause of blindness in working age adults
- Is a leading cause of kidney failure and dialysis
- Increases the risk of heart attacks and stroke by up to four times
- Is a major cause of limb amputations
- Affects mental health as well as physical health. Depression, anxiety and distress occur in more than 30% of all people with diabetes

Why is diabetes increasing?

Recent data shows one in twenty Australians currently live with diabetes^{ii, iii} and all types of diabetes are increasing in prevalence.

The number of National Diabetes Services Scheme (NDSS) registrants shows^{iv}:

- Type 1 diabetes currently accounts for 9% of all diabetes in Australia
- Type 2 diabetes accounts for 86.8% of all diabetes
- Gestational diabetes 3.4%
- Other diabetes is 0.8%

Note that the number of registrants is an underestimate of the number of people living with diabetes in Australia.

Type 2 diabetes is increasing at the fastest rate. There are large numbers of people with silent, undiagnosed type 2 diabetes which may be damaging their bodies. An estimated 2 million Australians are at high risk of developing type 2 diabetes and are already showing early signs of the condition.

Type 2 diabetes is one of the major consequences of the modern lifestyle. The combination of easy availability of high energy (kilojoule) foods at cheap prices, combined with more sedentary work and less physical activity, means most populations are developing more type 2 diabetes.

Genes also play a part with higher risk of type 2 diabetes in Chinese, South Asian, Indian, Pacific Islander and [Aboriginal and Torres Strait Islander](#) populations.

[i] The Royal Australian College of General Practitioners. Management of type 2 diabetes: A handbook for general practice. East Melbourne, Vic: RACGP, 2020

[ii] *Diabetes, 2020-21 financial year* | Australian Bureau of Statistics. (2022, March 21). www.abs.gov.au. <https://www.abs.gov.au/statistics/health/health-conditions-and-risks/diabetes/latest-release#diabetes-prevalence>

[iii] Australian Institute of Health and Welfare. (2022, July 13). *Diabetes: Australian facts, Summary*. Australian Institute of Health and Welfare <https://www.aihw.gov.au/reports/diabetes/diabetes/contents/summary>

[iv] *StackPath*. (n.d.). Map.ndss.com.au. <https://map.ndss.com.au/>, accessed 14/9/2022